

Hotel Booking Data Analysis Using Microsoft Excel

1. Project Overview

This assignment is based on the Hotel Booking Demand dataset. The main purpose of this assignment was to use Microsoft Excel to understand the hotel booking data and identify useful patterns, trends, and observations.

The analysis was performed using basic Excel data-analysis techniques such as data cleaning, formulas, Excel Tables, PivotTables, and charts.

The analysis focused on hotel bookings, cancellations, customer information, market segments, booking patterns, room information, and reservation details.

2. Objective

The main objectives of this assignment were:

- To understand the structure of the hotel booking dataset.
 - To clean and prepare the data for analysis.
 - To practice Excel formulas on a real-world dataset.
 - To analyze hotel booking and cancellation patterns.
 - To compare City Hotel and Resort Hotel.
 - To understand customer and market segments.
 - To analyze booking patterns across different months.
 - To create PivotTables for summarizing the data.
 - To create charts for better visualization.
 - To identify useful insights and findings from the dataset.
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3. Dataset Description

The Hotel Booking Demand dataset contains information about hotel reservations and customer bookings.

The dataset contains **119,390 booking records** and originally contains **32 columns**.

Some important columns used during the analysis include:

Column	Description
hotel	Type of hotel
is_canceled	Shows whether the booking was cancelled
lead_time	Number of days between booking and arrival

Column	Description
arrival_date_year	Arrival year
arrival_date_month	Arrival month
adults	Number of adults
children	Number of children
babies	Number of babies
meal	Meal type selected
country	Customer country
market_segment	Market segment
customer_type	Type of customer
reserved_room_type	Reserved room type
assigned_room_type	Assigned room type
adr	Average Daily Rate
total_of_special_requests	Number of special requests
reservation_status	Final reservation status

4. Data Cleaning and Preparation

The original dataset was first copied into a separate worksheet named **clean_data** so that the original data could be preserved.

The data was then converted into an Excel Table named **Hotel_Data**. This made it easier to use structured references in formulas and work with the dataset.

The data was checked for:

- Blank values
- Data types
- Duplicate or unusual values
- Numeric columns
- Categorical columns
- Cancellation values
- Missing information

The columns containing personal contact information such as name, email, phone number, and credit card information were not used for the analysis.

Some columns contained blank values, such as country, agent, company, and children. These were not removed unnecessarily because the blank values are part of the original dataset.

The value **Undefined** found in the meal column was treated as an existing category rather than a blank value.

5. Excel Formulas Used

Several Excel formulas were used to perform basic calculations and analysis.

The main formulas used were:

- COUNTA
- COUNTIF
- COUNTIFS
- SUM
- AVERAGE
- AVERAGEIF
- AVERAGEIFS
- IF
- IFERROR
- VLOOKUP
- HLOOKUP

These formulas helped calculate booking totals, cancellations, averages, and other measures.

Example calculations

Total Bookings

=COUNTA(Hotel_Data[hotel])

Cancelled Bookings

=COUNTIF(Hotel_Data[is_canceled],1)

Not Cancelled

=COUNTIF(Hotel_Data[is_canceled],0)

Cancellation Rate

=Cancelled Bookings / Total Bookings

Average Lead Time

=AVERAGE(Hotel_Data[lead_time])

Average ADR

=AVERAGE(Hotel_Data[adr])

These calculations were used as part of the summary analysis.

6. Summary Analysis

The summary analysis provided an overall view of the dataset.

Metric	Result
Total Bookings	119,390
Cancelled Bookings	44,224
Not Cancelled	75,166
Cancellation Rate	37.04%
Average Lead Time	Approximately 104 days
Average ADR	Approximately 101.83
Average Weekend Nights	Approximately 0.93
Average Week Nights	Approximately 2.50

The summary shows that a significant portion of bookings were cancelled, while the majority were not cancelled.

7. Hotel Analysis

The hotel category was analyzed using PivotTables and Excel formulas.

Hotel	Total Bookings	Cancelled	Not Cancelled	Cancellation %
City Hotel	79,330	33,102	46,228	41.73%
Resort Hotel	40,060	11,122	28,938	27.76%

The analysis shows that City Hotel had more bookings than Resort Hotel.

The cancellation percentage was also different between the two hotel types. City Hotel had a higher cancellation percentage in this dataset.

Average ADR was also calculated for both hotel types to compare their average daily rates.

8. Customer Analysis

Customer-related information was analyzed using PivotTables and formulas.

The analysis included:

- Repeat and non-repeat guests
- Number of adults
- Number of children
- Number of babies
- Customer types
- Special requests

The dataset contains a much larger number of non-repeat guests compared with repeat guests.

Customer type analysis was also performed using a PivotTable to understand the distribution of different customer categories.

9. Market Segment Analysis

The market_segment column was analyzed using a PivotTable.

The PivotTable was created with:

- **Rows:** Market Segment
- **Values:** Count of bookings

This helped identify which market segments contributed the largest number of bookings.

The analysis showed that **Online TA** was one of the major market segments in the dataset.

10. Monthly Booking Analysis

The arrival month was analyzed using a PivotTable.

The analysis included:

- Arrival month
- Number of bookings

A chart was created to make the monthly booking pattern easier to understand.

The monthly analysis helped identify months with relatively higher and lower booking volumes.

11. Reservation Status Analysis

The reservation_status column was analyzed using a PivotTable.

The main reservation statuses included:

- Check-Out
- Canceled
- No-Show

The analysis showed:

Reservation Status Bookings

Check-Out	75,166
Canceled	43,017
No-Show	1,207

This analysis helped understand the final status of reservations in the dataset.

12. Room Analysis

Room information was also analyzed using PivotTables.

The analysis included:

- Reserved room type
- Assigned room type

The number of bookings for different room types was compared to understand which room categories were more frequently reserved.

This also helped identify differences between the room originally reserved and the room eventually assigned.

13. Visualizations

Charts were created using the summarized Excel data and PivotTables.

The charts were used to make the analysis easier to understand visually.

The visualizations included comparisons related to:

- Hotel bookings
- Cancellation patterns
- Monthly bookings
- Market segments
- Customer categories
- Reservation status

Charts were useful for identifying differences between categories without having to examine the complete dataset row by row.

14. Key Insights and Findings

Based on the Excel analysis, the following observations were identified:

1. The dataset contains **119,390 hotel booking records**.
 2. Around **37% of bookings were cancelled**, showing that cancellation is an important factor in the dataset.
 3. **City Hotel** had more total bookings than Resort Hotel.
 4. The cancellation percentage for **City Hotel was higher than Resort Hotel** in this dataset.
 5. The average lead time was approximately **104 days**, meaning bookings were often made well before the arrival date.
 6. The average daily rate was approximately **101.83** across the complete dataset.
 7. Non-repeat guests represented the large majority of bookings, while repeat guests represented a relatively small portion.
 8. **Online TA** was a major market segment by booking volume.
 9. The monthly analysis showed differences in booking volume between arrival months.
 10. The reservation-status analysis showed that most reservations were completed with a **Check-Out** status, while cancellations and no-shows represented smaller portions.
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15. Conclusion

This assignment helped me understand how Microsoft Excel can be used to analyze a real-world dataset.

I started by understanding and preparing the hotel booking data. After that, I used Excel Tables, formulas, PivotTables, and charts to summarize the information.

The analysis helped me understand booking volumes, cancellation patterns, hotel differences, customer types, market segments, room types, and reservation statuses.

Overall, this assignment provided practical experience in using Excel for basic data analysis and converting raw data into meaningful information.

16. Tools Used

Software: Microsoft Excel

Techniques Used:

- Data Cleaning

- Excel Tables
- Excel Formulas
- Filtering and Sorting
- PivotTables
- Charts
- Basic Data Analysis
- Insight Generation